

## Practical Exercise 1

## Questions for discussion

1. Briefly explain, the main features of High available cluster?
2. Mr. P claims that Hadoop data storage is an example of data lake repository. Are you agree? Why?
3. Use Microsoft Excel, extract the **ABC CAPITAL PRIMARY SCHOOL** standard seven results data for the year 2022,2023, and 2024 from the NECTA website, then prepare a simple data warehouse based on dimensions of sex, number of students, year, math grade, and number of math grades. Save the data warehouse in a workbook called “**abc\_school.xlsx**”. **Hint: ABC CAPITAL** is one among the primary school of Dar es salaam region located at the Dar es salaam City Council.
4. Use the relational table data below to write the corresponding JSON file. **Hint:** use notepad and save the name of the file as **qn3.json**

Purchase table

| Transaction ID | Customer ID | Product ID | Purchase date |
|----------------|-------------|------------|---------------|
| 1112           | 24221       | 8977       | 03-22-2010    |
| 1113           | 24222       | 8978       | 03-22-2010    |
| 1114           | 24223       | 8979       | 03-22-2010    |

| Customer ID | Customer | Address          |
|-------------|----------|------------------|
| 24221       | Bob      | 123 East street  |
| 24222       | Alice    | 223 Main street  |
| 24223       | Martha   | 465 North street |

Customer table

| Product ID | Name   | Price |
|------------|--------|-------|
| 8977       | Banana | .79   |
| 8978       | TV     | 400   |
| 8979       | Watch  | 50    |

Product table